

Clinical Study Summary Report (CSSR)

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1. Administrative & Study Identification

Full Study Title: A Study to Evaluate the Efficacy, Safety, and Tolerability of MYK-224 in Participants With Symptomatic Obstructive Hypertrophic Cardiomyopathy **Short Title/Acronym:** MERCUTIO **Protocol Number:** CV029-009 / BMS-986435 **NCT Number:** NCT05556343 **Sponsor Name:** Bristol-Myers Squibb **Investigational Product:** MYK-224 (BMS-986435) **Phase of Development:** Phase 2a **Medical Monitor:** Bristol-Myers Squibb Medical Monitor

2. Study Synopsis & Rationale

2.1 Study Objective **Primary Objective:** To characterize the safety, tolerability, efficacy, pharmacokinetics (PK), and pharmacodynamics (PD) of MYK-224 in participants with obstructive Hypertrophic Cardiomyopathy (oHCM). **Secondary Objectives:** To evaluate the change in left ventricular outflow tract (LVOT) peak gradient (post-exercise, resting, and Valsalva) from baseline to the end of treatment, and assess the proportion of participants achieving target resting and Valsalva LVOT peak gradients.

2.2 Background & Rationale To investigate the clinical profile of MYK-224, a next-generation cardiac myosin inhibitor, as a treatment to alleviate symptoms and reduce the left ventricular outflow tract obstruction in patients suffering from symptomatic obstructive Hypertrophic Cardiomyopathy (oHCM).

3. Study Design & Methodology

3.1 Design Framework **Study Type:** Interventional **Control Type:** Open-label / Standard-of-care combination (e.g., Cohort 2 standard-of-care includes calcium channel blocker or disopyramide) **Blinding:** Open-label **Randomization:** N/A (Pilot Study / Cohort Design)

3.2 Planned Enrollment & Duration **Target \$N\$:** Not explicitly defined (Phase 2a Pilot) **Number of Sites:** Multicenter global study **Estimated Study Start:** 18-Jan-2023 (Actual) **Estimated Primary Completion:** 19-Aug-2024 (Actual) / Study Completion: 27-Feb-2025 (Actual)

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Men or women with unexplained left ventricular (LV) hypertrophy with nondilated ventricular chambers, and maximal LV wall thickness ≥ 15 mm (or ≥ 13 mm with a positive family history or known disease-causing mutation). 2. LVOT peak gradient ≥ 50 mm Hg at rest, or ≥ 30 mm Hg at rest and ≥ 50 mm Hg with Valsalva maneuver (confirmed by core laboratory interpretation). 3. Resting LVEF $\geq 60\%$ and New York Heart Association (NYHA) functional class II or III symptoms at screening.

4.2 Exclusion Criteria 1. History of syncope or sustained ventricular tachyarrhythmia within 6 months prior to screening. 2. Known infiltrative or storage disorder causing cardiac hypertrophy mimicking HCM (e.g., Fabry disease, amyloidosis, Noonan syndrome). 3. Prior treatment with mavacamten or aficamten (within 4 months of the Screening visit).

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: MYK-224 (BMS-986435) administered at specified doses on specified days. **Route of Administration:** Oral. **Duration of Treatment:** Up to 45 weeks for primary gradient assessment endpoints, with safety monitored up to 53 weeks. Includes an optional open-label extension period.

5.2 Concomitant & Prohibited Therapy Permitted: Standard-of-care therapies consisting of either a calcium channel blocker or disopyramide (in combination with beta-blockers or calcium channel blockers). **Prohibited:** Prior treatment with cardiotoxic agents such as anthracyclines (e.g., doxorubicin).

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Day -28 to 0	Informed Consent, Medical History, Echocardiography (LVEF, LVOT peak gradient), ECG
Baseline	Day 0	Enrolment, First Dose of MYK-224
Interim	Weeks 1 to 45	PK/PD assessments, Safety Labs, TTE (Echocardiography for LVOT gradient evaluation)
End of Study	Week 53	Final Safety Assessment, AE and vital sign abnormality resolution

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Incidence of adverse events (AEs), arrhythmias, vital sign abnormalities, physical examination abnormalities, ECG abnormalities, TTE abnormalities,

and clinical laboratory abnormalities. **Measurement Method:** Clinical, laboratory, and echocardiogram (TTE) assessments over a time frame of up to 53 weeks.

7.2 Secondary & Exploratory Endpoints 1. Change in left ventricular outflow tract (LVOT) peak gradient (post-exercise, resting, and Valsalva) from baseline to the end of treatment (up to 45 weeks). 2. Proportion of participants achieving a resting LVOT peak gradient of < 30 mm Hg and a Valsalva LVOT peak gradient < 50 mm Hg at end of treatment. 3. Concentration-response relationship between MYK-224 PK and LVOT peak gradients, as well as echocardiographic parameters of systolic and diastolic function.

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Standard continuous monitoring encompassing patient visits, ECG evaluation, and clinical laboratory tests (collected up to 53 weeks). **Serious Adverse Events (SAEs):** Monitored up to 53 weeks, specifically including incidence of appropriate implantable cardioverter-defibrillator (ICD) therapy and resuscitated cardiac arrest. **Data Monitoring Committee (DMC):** Internal safety monitoring overseen by the Medical Monitor and Investigator.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Safety Set for all primary AE/SAE evaluations and PK/PD population sets for concentration-response relationships. **Sample Size Justification:** Sample size aligned with a Phase 2a open-label pilot design to detect primary safety, tolerability, and initial efficacy profiles. **Handling of Missing Data:** Standard protocol-defined multiple imputation and last observation carried forward methods where applicable.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Centralized core laboratory interpretations are required for critical assessments such as echocardiography (LVEF, LVOT peak gradients) to ensure consistency. **Audit Trail:** eCRF logging, site audits, and standard quality audit mechanisms.

11. Study Identifiers & Governance

Primary ID: CV029-009 **Registry Number:** NCT05556343 **Sponsor/Lead Organization:** Bristol-Myers Squibb **Study Title:** MERCUTIO **Official**

Regulatory Title: A Phase 2a, Open-label, Pilot Study to Evaluate Efficacy, Pharmacokinetics, Pharmacodynamics, Safety, and Tolerability of MYK-224 in Participants With Symptomatic Hypertrophic Cardiomyopathy and Left Ventricular Outflow Tract Obstruction (MERCUTIO)

12. Clinical Framework (Study Specific)

Primary Condition: Symptomatic Obstructive Hypertrophic Cardiomyopathy (oHCM) **Diagnosis Threshold:** Maximal LV wall thickness ≥ 15 mm; LVOT peak gradient ≥ 50 mm Hg at rest **Medical Methodology:** Cardiac Myosin Inhibition **Optimization Target:** Reduction of LVOT peak gradient and symptom alleviation

13. Study Procedures & Methodology

Management Pathway: Targeted cardiac myosin inhibitor administration alongside monitored standard-of-care. **Education Protocol (HCP):** Routine site initiation and continuous investigator training on protocol adherence and core lab imaging standards. **Education Protocol (Patient):** Informed consent, stress-testing expectations, and cardiac monitoring self-awareness. **Quality Audit Mechanism:** Centralized echocardiography core laboratory interpretation.

14. Observation & Follow-Up Schedule

Total Duration: Up to 53 weeks (45 weeks evaluation + safety follow-up) **Onsite Visit Cadence:** Pre-defined periodic assessments for TTE, stress testing, and PK evaluations. **Remote Monitoring Frequency:** Standard AE follow-up tele-visits as outlined in the protocol. **Checklist Review Interval:** Routine continuous assessment up to Week 53.

15. Sign-off & Approval

Role	Name	Signature	Date	:---	:---	:---	:---
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]		Lead		
Statistician	[Name]	_____	[DD-MMM-YYYY]				